Flow in Wind Tunnels

AA 504 (Compressible Fluid Mechanics) Autumn 2026, UW

Assignment: Homework #3

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Problem 1. Waves in Fluids

Problem Statement

Based off of the movie "Waves in Fluids" [1], write one paragraph on the most surprising aspect of the movie.

Statement

Waves in fluids are traveling disturbances that transfer energy and momentum through a fluid, even though the fluid particles themselves may move only a short distance as the wave passes. One surprising feature is that a water wave, driven by gravity, and a sound wave which even though is driven by a different mechanism - gas compressibility, still follow many of the same principles.

What makes wave motion surprising is that the disturbance can travel far faster and farther than the individual fluid particles it passes through. The video also shows that waves can change shape as they travel: compression waves tend to steepen, while expansion waves spread out. In water, a steep wave may break into a turbulent surge; in a gas, a steep compression can form a remarkably thin shock wave.

Problem 2. Rayleigh Flow

Problem Statement

Steady flow of air enters a tube of diameter D at an initial Mach number of 0.4. Assume no friction. Determine the amount of heat necessary to thermally choke the flow and the exit total pressure.

Analysis

Assumptions:

- No friction (Rayleigh flow)
- Inlet static standard atmospheric conditions:
  - $P_1 = 1\,atm = 101.325\,kPa$
  - $T_1 = 300\,K$
- Inlet *stagnation* conditions:
  - At $M_1 = 0.4$:
  - $T_{t1} = T_1\left(1 + \frac{\gamma - 1}{2} M_1^2\right) = 309.6\,K$
  - $p_{t1} = p_1\left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\gamma/(\gamma-1)}$
    $= 113.1\,\text{kPa}$

Begin with first principles; **conservation of energy.**

Heat added + Work done = change in Energy

$q + w = \Delta E$

- $w = 0$; there is no (shaft) work being done on flow by any kind of device (turbine, pump, etc.)

$q = \Delta h_{total}$; total enthalpy

- $\Delta h_{total} = h_2 + \frac{u_2^2}{2} - h_1 - \frac{u_1^2}{2}$
- total enthalpy is the change in enthalpy measured at two equilibrium end points (station 2 and station 1)

$$\frac{q^*}{c_p T_{t1}} = \frac{T_t^* - T_{t1}}{T_{t1}} \quad \rightarrow \quad \frac{q^*}{h_{t1}} = \frac{T_t^*}{T_{t1}} - 1$$

From the Rayleigh flow chart, T_t/T_t^* curve:

- $\frac{T_t}{T_t^*} \cong 0.52$

$$\rightarrow \frac{q^*}{h_{t1}} = \frac{T_t^*}{T_{t1}} - 1 = 0.92$$

From the energy equation for a perfect gas, for adiabatic flow, we obtain the following temperature ratio - Mach # relation:

$$\frac{T_{t1}}{T_1} = 1 + \frac{\gamma - 1}{2} M^2 \quad (2.30) \quad [3]$$

- At Mach = 0.4: $\frac{T_{t1}}{T_1} = 1.032$

$$\frac{q^*}{h_1} = \frac{q^*}{c_p T_{t1}} \frac{T_{t1}}{T_1} = (0.92)(1.032) = 0.95$$

Air: $c_p = 7/29 \ \frac{cal}{g \cdot K}$

Heat to Thermally Choke the Flow

$$q^* = h_1 \frac{q^*}{c_p T_{t1}} \frac{T_{t1}}{T_1} = c_p T_1 \frac{q^*}{c_p T_{t1}} \frac{T_{t1}}{T_1}$$

- Note: h_1 is the *static* enthalpy at the inlet.

At $T_1 = 300\,K$ (*static inlet temperature*) $\rightarrow$ **q* = 68.74 cal/g**

$\rightarrow$ This is the amount of heat, in thermochemical calories per gram, needed to thermally choke the flow.

Exit Total Pressure

From the Rayleigh flow chart:

- $\frac{p_{t1}}{p^*} \cong 1.16$

$$p_{t2} = p_t^* = \frac{113.1 kPa}{1.16} = \mathbf{97.5\ kPa}$$

$\rightarrow$ This is the pressure at the choke point.

Pressure drop from the inlet stagnation pressure:

$$\% \frac{\Delta p_t}{p_{t1}} = \frac{97.5 - 113.1}{113.1} 100\,\% = -13.79\,\%$$

→ The pressure drop is 13.79% from the inlet stagnation pressure.

Problem 3. Fanno Flow

Problem Statement

Steady flow of air enters a tube of diameter D at an initial Mach number of 0.4. Assume an adiabatic wall. Determine the length for frictional choking, the exit total pressure, and the exit total temperature. The skin friction coefficient is 0.005.

Analysis

Assumptions:

- Adiabatic wall, $q = 0$
- Skin friction coefficient, $c_f = 0.005$
- Inlet static standard atmospheric conditions:
  - $P_1 = 1\ atm = 101.325\ kPa$
  - $T_1 = 300\ K$
- Inlet *stagnation* conditions:
  - At $M_1 = 0.4$:
  - $T_{t1} = T_1\left(1 + \frac{\gamma - 1}{2} M_1^2\right) = 309.6\ K$
  - $p_{t1} = p_1\left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\gamma/(\gamma-1)}$
    $= 113.1\ \text{kPa}$

From the Fanno Flow chart at $M_1 = 0.4$:

- $\frac{4C_f L^*}{D} \cong 2.2$
- $\frac{p_{t1}}{p_t^*} \cong 1.58$
- $\frac{\rho}{\rho *} \cong 2.62$

Length for Frictional Chocking

$$L^* = \frac{2.2D}{4c_f} \approx 110D$$

→ Choking occurs after approximately 110 tube diameters.

Exit Total Pressure

$$p_t^* = p_{t2} = \frac{p_{t1}}{1.58} = \mathbf{71.6\,kPa}$$

→ The pressure at the choke point is about ~63% of the inlet *stagnation* pressure.

Exit Total Temperature

→ For steady adiabatic flow with no shaft work:

$$h_t = h + \frac{u^2}{2} = const.$$

Consequently: $T_{t2} = T_{t1} = 309.6\,K$

→ Friction reduces the total pressure, while the total temperature stays constant because no energy enters or leaves the flow as heat or work.

References

[1] YouTube '14. Waves in Fluids'.

[2] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.

[3] Liepmann, H. W., & Roshko, A. (2001). *Elements of gasdynamics*. Dover Publications. (Original work published 1957)